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PHANTOM POSITIONS

IDENTIFYING GHOST JOB POSTINGS IN ONLINE LISTINGS - A PROPOSAL

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Synopsis and Problem Statement

The proliferation of online job platforms has created significant opportunities for both employers and job seekers; however, it has also introduced a vector for fraud. Fraudulent job postings, which are listings designed to extract personal information, solicit fees, or otherwise deceive applicants, represent a growing threat to vulnerable populations, including recent graduates, career changers, and the unemployed. The consequences of engaging with such postings range from financial loss to identity theft.

To help address the **problem of fraudulent job listings and to support automated detection of such**, this project, *Phantom Positions*, will apply data science techniques, specifically supervised classification and text mining, to a publicly available labeled dataset of real and fake job postings. We will evaluate multiple classification approaches across three feature conditions and identify the best-performing configuration as the basis for a prospective screening tool.

Literature Review

Prior work on this problem has largely converged on the Employment Scam Aegean Dataset (EMSCAD). Vidros et al. (2017) [1] introduced the dataset and established the first classification baselines using bag of words features across six classifiers, with Random Forest achieving an F-measure of 0.912 on a balanced subsample. Critically, when tested on the full dataset, fraudulent class precision dropped to 0.282, directly motivating our use of F1-score and resampling strategies. Amaar et al. (2022) [2] subsequently demonstrated that combining TF-IDF text features with structured metadata outperforms either feature type alone. Bhatia and Meena (2023) [3] applied TF-IDF with SMOTE and ADASYN oversampling to address class imbalance, providing methodological precedent for our approach. Pillai (2023) [4] represents the current state of the art, employing a Bi-LSTM deep learning model, establishing a performance ceiling against which our interpretable classification models can be contextualized.

Dataset Description

This project uses the "Real or Fake Job Posting Prediction" dataset, sourced from the EMSCAD [1] and available publicly on Kaggle[1]. The dataset contains 17,880 job postings labeled for binary classification (legitimate or fraudulent) across 18 features. Structured metadata features include binary indicators (company logo presence, screening questions, telecommuting status) and nominal fields (employment type, required experience, required education, industry, and function). Free-text features include the job title, company profile, job description, requirements, and benefits.

The dataset is highly imbalanced at approximately 95% legitimate and 5% fraudulent, which makes standard accuracy an inadequate evaluation metric. A classifier predicting every posting as legitimate would achieve 95% accuracy while

[1] https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction/data

detecting no fraud whatsoever. F1-score is therefore selected as the primary metric, as it accounts for both precision and recall and is sensitive to performance on the minority class.

Proposed Approach and Programming Tools

This project will be implemented in Python using pandas, scikit-learn, matplotlib, and seaborn within a Jupyter Notebook environment.

Preprocessing and Feature Engineering

Free-text fields will be preprocessed using tokenization, stopword removal, and stemming, then represented as TF-IDF weighted vectors [2, 3]. Structured metadata features will be encoded using binary and one-hot encoding and combined with the TF-IDF representation to form a unified feature matrix for the combined condition. SMOTE will be applied to address class imbalance, consistent with prior work [3].

Classification Models

We will evaluate the following two classification approaches, both of which are covered in the course, across three feature conditions: text features only, metadata features only, and combined text and metadata features.

- *Support Vector Machine (SVM)*: SVMs are well-suited to high-dimensional, sparse feature spaces such as those produced by TF-IDF vectorization, and are covered as an advanced classification technique in the course.
- *Random Forest*: An ensemble method that handles mixed feature types (text and metadata) and provides feature importance scores, supporting the course objective of visualizing relationships between variables.

Evaluation

Both models will be evaluated using F1-score as the primary metric, with precision, recall, and confusion matrices reported for completeness. Where applicable, ROC curves will be plotted to illustrate the tradeoff between true positive and false positive rates across classification thresholds.

Anticipated Outcomes and Impact

We anticipate that combined text and metadata features will outperform either feature type in isolation, with Random Forest performing strongest on the combined condition and SVM on the text-only condition. The findings are framed in the context of a prospective platform-side screening tool, where both false negatives (missed fraud) and false positives (flagged legitimate postings) carry real costs, reinforcing F1-score as the appropriate metric.

The novel contribution of this project is a systematic, side-by-side evaluation of three feature conditions (text only, metadata only, and combined) across both classifiers, applied to the full imbalanced dataset. This explicit isolation of feature type contributions has not been the primary focus of prior work on this dataset and is expected to clarify which feature types drive classification performance in the fraudulent job detection task.

References

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